



UNIVERSITÄT
LEIPZIG

Asymmetric Numeral System

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Outline

1. Introduction
2. Entropy Coding
3. Asymmetric Numeral System
4. rANS
5. Streaming-rANS
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Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

Entropy Coding

- The theoretical limit of compression is given by Shannon entropy, which measures the minimum average number of bits needed to encode a symbol from a data source.
- Lossless compression algorithms aim to approach this limit but cannot go below it.
- The entropy depends on the probability distribution of the symbols, with more skewed distributions being more compressible.
- Shannon Entropy ^[1]

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

- Shannon's Source Coding Theorem ^[1]

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding ^[4].

Basic Idea

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- Representation of information
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(represents an entire message as a single number, similar to Arithmetic Coding) → example: representation of {2,0,2,5,1,8}
- Symmetric or Asymmetric
- rANS and tANS

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- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]

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- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
e.g. splits natural numbers into subsets of even/odd numbers [2]

Symbol spread function (Exa. 1)

$$S = \{0, 1\}$$

Exa. 1: Standard Binary System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
s=0																			
s=1																			

Fig.1

Symbol spread function (Exa. 2)

$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$x' = 10x + s$$

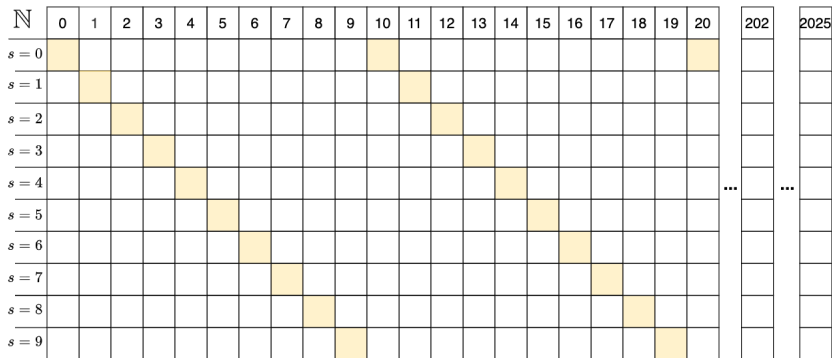


Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

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- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$x_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$x_{\text{init}} = 0$$

$$x_1 = x_{\text{init}} \times 10 + 2 = 2$$

$$x_2 = x_1 \times 10 + 0 = 20$$

$$x_3 = x_2 \times 10 + 2 = 202$$

$$x_{\text{out}} = x_3 \times 10 + 5 = 2025$$

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$$x_{i+1} = x_i \times 10 + s_i$$

- x' is the x -th appearance of symbol s

$$x' = C(s, x) = x \cdot 10 + s$$

Representation of Information (Exa. 2)

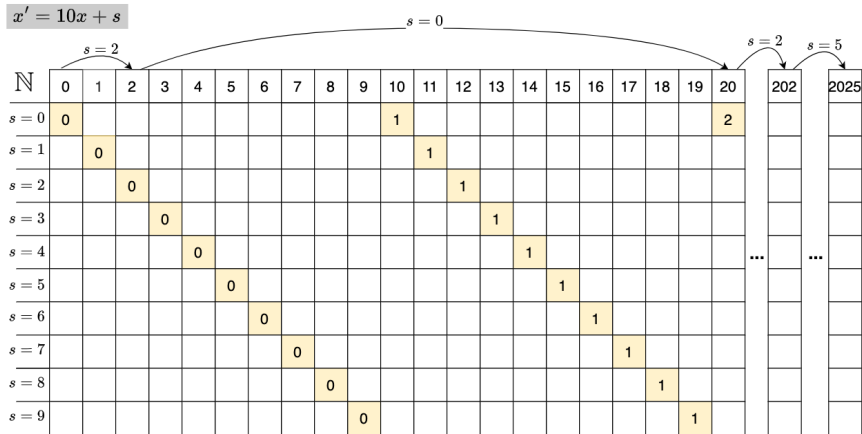


Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

$X' = 2X + s$

$s=1$ $s=0$ $s=0$ $s=1$

x'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x																			
$s=0$	0		1		2		3		4		5		6		7		8		9
$s=1$		0		1		2		3		4		5		6		7		8	

Fig.4

Asymmetric Numeral System | Symmetric or Asymmetric |

Exa. 1: Standard Binary System

$$X' = 2X + s$$

The diagram illustrates the bit-level operation of the shift operation $X' = 2X + s$. It shows a 19-bit register X' and a 4-bit register X . The register X' is divided into four 4-bit segments. The first segment (bits 0-3) is labeled $s=1$, the second (bits 4-7) is labeled $s=0$, the third (bits 8-11) is labeled $s=0$, and the fourth (bits 12-15) is labeled $s=1$. The register X is divided into two 4-bit segments. The first segment (bits 0-3) is labeled $s=0$, and the second (bits 4-7) is labeled $s=1$. The diagram shows the shift operation $X' = 2X + s$, where the value of s is shifted into the least significant bits of X' .

Exa. 2: Standard Decimal System

$$X' = 10X + s$$

[illegible]

Fig.6

Exa. 3: Asymmetric Binary System

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$

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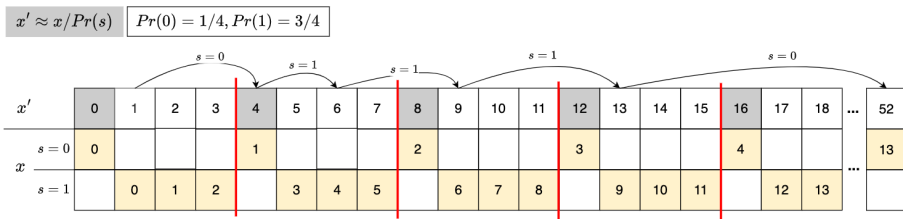


Fig.7

We refine even/odd number by changing their densities

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

tANS

— d

Performance

— f

rANS

rANS

- "r" = "range": similar to range coding (RC)
- symbol spread function
- representation of information
- asymmetric

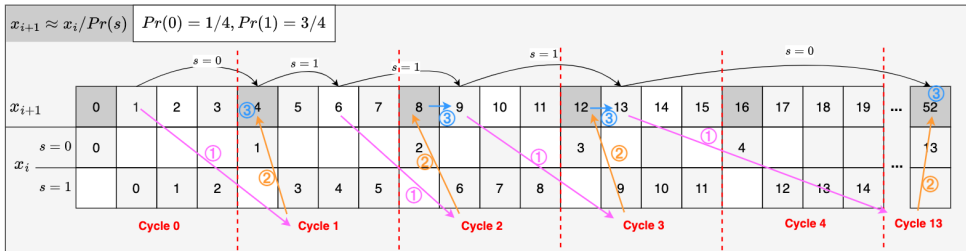
Exa.3

How to encode 101110? What is the encoding function $C(s, x)$?

$$x' = C(s, x) \approx x/p_s$$

Some notations

- $\mathcal{A} = \{s_1, s_2, \dots, s_k\}$ is the alphabet set
- s is the symbol, if it is used in subscript: s is the current symbol, $s - 1$ is the previous symbol, $s + 1$ is the next symbol
- F_s is the absolute frequency of symbol s
- $m = \sum_{s \in \mathcal{A}} F_s$ is the total frequency
- $c_s = \sum_{s \in \mathcal{A}} F_{s-1}$ is the cumulative frequency (c_{s+1} is the cumulative frequency of the next symbol)
- $p_s = F_s/m$ is the probability of symbol s
- x_i is the current state (next state: x_{i+1} , previous state: x_{i-1})



Given current state x_i and next symbol s . How to find next state x_{i+1} , which is the x_i -th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$		General
	$s = 0$	$s = 1$	
Step ①: Find cycle number k	$k = \lfloor x_i / 1 \rfloor$	$k = \lfloor x_i / 3 \rfloor$	$k = \lfloor x_i / F_s \rfloor$
Step ②: Navigate to the beginning (first number) of that cycle	$4 \cdot k$	$4 \cdot k$	$m \cdot k$
Step ③: Find the position of the symbol s in that cycle	part 1: first of one + 0	part 1: + mod($x_i, 3$)	part 1: + mod(x_i, F_s)
- part 1: position of the symbol s in the sequence of the same symbol - part 2: how many other symbols before the symbol s	part 2: no other symbols + 0	part 2: only one other symbol (0) and with one of that symbol in one cycle + 1	part 2: only one other symbol (0) and with one of that symbol in one cycle + c_s
	$x_{i+1} = m \cdot k$	$x_{i+1} = 4 \cdot \lfloor x_i / 3 \rfloor + \text{mod}(x_i, 3) + 1$	$x_{i+1} = m \cdot \lfloor x_i / F_s \rfloor + \text{mod}(x_i, F_s) + c_s$

$$x_{i+1} = C(s, x_i) = m \cdot \lfloor x_i / F_s \rfloor + \text{mod}(x_i, F_s) + c_s$$

Fig.8

Encoding Function [2] [5]:

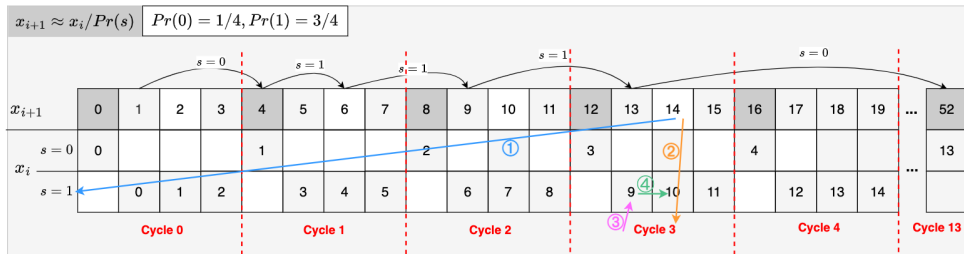
$$x' = m \cdot \lfloor x/F_s \rfloor + \text{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$
- **Absolute frequency of symbol s :** F_s
- **Total frequency:** $m = \sum_{i=1}^k F_i$
- **Probability of symbol s :** $p_s = \frac{F_s}{m}$
- **Cumulative frequency before s :** $c_s := F_0 + F_1 + \dots + F_{s-1}$

Encoding Function (binary) ^[2] : maybe no need to put this slide, let's see

$$x' = C(s, x) = \lfloor x/F_s \rfloor \ll n + \text{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$
- **Absolute frequency of symbol s :** F_s
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- **Probability of symbol s :** $p_s = \frac{F_s}{m}$
- **Cumulative frequency before s :** $c_s := F_0 + F_1 + \dots + F_{s-1}$
- $\ll n$ is a left shift by n bits



Given current state x_i . How to find previous state x_{i-1} , where x_i is the x_{i-1} -th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$		General
	$s = 0$	$s = 1$	
Step ① : Find the symbol s for the state x_i	if $x_i \bmod 4 = 0$	otherwise or if $x_i \bmod 4 = 1, 2, 3$	s such that $c_s \leq \text{mod}(x_i, m) < c_{s+1}$
Step ②: Find the cycle k of the current state x_i	$k = x_i / 4$	$k = \lfloor x_i / 4 \rfloor$	$k = \lfloor x_i / m \rfloor$
Step ③: Find the first number of that symbol s in that cycle k	$x_i / 4 \cdot 1$	$\lfloor x_i / 4 \rfloor \cdot 3$	$\lfloor x_i / m \rfloor \cdot F_s$
Step ④: Allocate to the exact position	$x_{i-1} = x_i / 4 \cdot 1 + 0 - 0$	$x_{i-1} = \lfloor x_i / 4 \rfloor \cdot 3 + 2 - 1$	$x_{i+1} = \lfloor x_i / m \rfloor \cdot F_s + \text{mod}(x_i, F_s) - c_s$

1. s such that $c_s \leq \text{mod}(x_1, m) < c_{s+1}$
2. $x_{i-1} = D(x_i) = \lfloor x_i / m \rfloor \cdot F_s + \text{mod}(x_i, F_s) - c_s$

Fig.9

Decoding Function (binary):

$$D(x) = (s, F_s \cdot (x \gg n) + (x \& \textit{mask}) - c_s)^{[2]}$$

```
1 def rANS_encoder(symbol_counts, s_input):
2     # compute cumulative frequencies
3     cumul_counts = []
4     sum_counts = 0
5     for count in symbol_counts:
6         cumul_counts.append(sum_counts)
7         sum_counts += count
8     state = 0
9     output = []
10    for s in s_input:
11        Fs = symbol_counts[s]
12        Cs = cumul_counts[s]
13        state = (state // Fs) * sum_counts + Cs + (state % Fs)
14        output.append((s, state))
15
16    L_avg = math.ceil(math.log(state) / math.log(2.0)) / len(s_input)
17    return output, state, L_avg, entropy(symbol_counts)
```

Implemenation in Dash app with interactive visualization

```
1     def rANS_decoder(symbol_counts, num_symbols, state):
2         cumul_counts = [0] # Start with 0 as the first cumulative frequency
3         total_frequency = sum(symbol_counts)
4         for count in symbol_counts:
5             cumul_counts.append(cumul_counts[-1] + count)
6         def c_inv(y):
7             return bisect_right(cumul_counts, y) - 1
8         output = [] # List to store decoded symbols
9         for _ in range(num_symbols):
10             slot = state % total_frequency # Find the slot in the cumulative
11                 ↪ frequency range
12             s = c_inv(slot) # Get the symbol corresponding to the
13                 ↪ slot
14             Fs = symbol_counts[s] # Frequency of the symbol
15             Cs = cumul_counts[s] # Cumulative count of the symbol
16             output.append((s, state)) # Append decoded symbol to the
17                 ↪ output
18             state = (state // total_frequency) * Fs + (slot - Cs) # Update the
19                 ↪ state
20         return output, state
```

Streaming-rANS

Streaming-rANS

To avoid using large number arithmetic, in practice stream variants are used: which enforce by renormalization: sending the least significant bits of x to or from the bitstream (usually L and b are powers of 2). ^[4]

Example of the problem //TODO: add example

- ANS (Asymmetric Numeral Systems) uses a single integer x , known as the “state,” to represent the encoded data.
- Over time, as more symbols are encoded, the state x can grow arbitrarily large. This can lead to:
 - Overflow: If the value exceeds the precision limits of your system (e.g., a 32-bit or 64-bit register).
 - Inefficient Encoding: A larger x means you need more memory or bits to represent it.
- Using the Cfunction from the previous section to encode succeeding symbols, we can encode a sequence into a very large natural number x , which can be stored as a length $\lfloor \lg(x) \rfloor + 1$ bit sequence. [2]

$[L, H]$

- How to determine L and H?

$[L, H]$

- How to determine L and H?

-

$[L, H]$

- How to determine L and H?

-

-

Method 1: ANS encoding step of symbol s from state x [2]

```

while  $x > \max X[s]$  do                                {fff}
    writeDigit(mod( $x, b$ ));  $x = \lfloor x/b \rfloor$             {fff}
end while
 $x = C(s, x)$                                            {fff}

```

Method 2: ANS decoding step from state x [2]

```

( $s, x$ ) =  $D(x)$                                            {fff}
useSymbol( $s$ )                                           {fff}
while  $x < L$  do                                       {fff}
     $x = b \cdot x + \text{readDigit}()$                        {fff}
end while

```

- Base decoding:
- $x_i = 2 * x_{i-1} + bit$
- Either x_i in $[L, H]$ or x_{i-1} in $[L, H]$. Assume x_{i-1} in $[L, H]$.
 - $x_{i-1} \geq L$
 - $x_i \geq 2 * L + bit \geq 2 * L$
 - x_i can't be in $[L, H]$
 - derive: $H \leq 2 * L - 1$

```
1     for i in range(10):  
2         print("Hello, World!")
```

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

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2. J. Duda, K. Tahboub, N. J. Gadgil and E. J. Delp, "The use of asymmetric numeral systems as an accurate replacement for Huffman coding," 2015 Picture Coding Symposium (PCS), Cairns, QLD, Australia, 2015, pp. 65-69, doi: 10.1109/PCS.2015.7170048.
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